

Second virial coefficients of normal alkanes, linear 1-alkanols (and water), alkyl ethers, and their mixtures

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Abstract

Earlier recommendations [J.H. Dymond et al., *Fluid Phase Equilibria* 27 (1986) 1–13; C. Tsonopoulos et al., *Pure Appl. Chem.* 61 (1989) 1387–1394.] are updated for C₄–C₈ alkanes and C₁–C₄ alkanols, and more recent values are presented for 1-hexanol. New recommendations are made for butane, on the basis of recent isochoric Burnett measurements. Preliminary recommendations are given for dimethyl, diethyl, and diisopropyl ethers, but nothing definitive can be said about other alkyl ethers due to the limited quantity and quality of the available data. Calculations with a correlation [C. Tsonopoulos, *AIChE J.* 20 (1974) 263–272] reexamine the dependence of the parameter *b* of alkanols and parameter *a* of alkyl ethers on their reduced dipole moment. Also included is the recent [C. Tsonopoulos and J.L. Heidman, *Fluid Phase Equilibria* 57 (1990) 261–276] analysis of the data for water and water/alkane binaries, along with new analyses of data for the binaries of water with methanol and ethanol. Only other relevant binary data that could be located were for *n*-hexane/ethanol and methanol/diethyl ether. Recommendations are given for the characteristic binary constant *k_{ij}* for all binaries of interest. In addition, recent results are provided for the critical properties of the industrially important alkyl ethers MTBE, ETBE, TAME, and TAEE, together with recommendations for calculating their *B* with the 1974 correlation. © 1997 Elsevier Science B.V.

Keywords: Equation of state; Method of calculation; Second virial coefficient; Alkanes; Alkanols; Alkyl ethers

1. Introduction

Dymond et al. [1] recommended second virial coefficient values for the first eight normal alkanes, on the basis of a critical review of all available experimental data. This 1986 review was expanded by

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Tsionopoulos et al. [2] by considering more recent data for butane, pentane, and hexane. In addition, preliminary recommendations were made for the C_1 – C_4 linear 1-alkanols. The 1989 review concluded with an analysis of second virial cross-coefficient data for alkane/alkane and alkane/1-alkanol binaries, and recommendations were made for the characteristic binary constant k_{ij} that is used in the correlation of Tsionopoulos [3].

This work updates the 1986 and 1989 reviews [1,2], primarily for butane and methanol, and expands the scope by considering the industrially important alkyl ethers. Several alkyl ethers are either used or are being considered as gasoline additives. The most prominent are

- MTBE (methyl *tert*-butyl ether)
- TAME (*tert*-amyl methyl ether)
- ETBE (ethyl *tert*-butyl ether)
- DIPE (diisopropyl ether)
- TAE (ethyl *tert*-amyl ether)

The properties of ethers are especially important in designing processes for their manufacture. For example, in the reaction of isobutylene with methanol to form MTBE, dimethyl ether and methyl *sec*-butyl ether are important byproducts.

This paper also examines water and, especially, water binaries, for which there are more plentiful second virial cross-coefficient data than for alkane/1-alkanol/alkyl ether binaries. As in the 1989 review [2], all the second virial cross-coefficient data are analyzed with the 1974 correlation [3] and recommendations are made for the characteristic binary constant k_{ij} , which corrects for the deviation of T_{cij} from the geometric mean.

2. Sources of data and their analysis

2.1. Data reduction

A variety of methods are used to measure second virial coefficients. The most common ones involve the measurement of density (or volume), the sonic velocity, or the Joule–Thomson coefficient. The p – v – T determinations are the most direct, but need to be corrected for adsorption effects, especially at low temperatures. Eubank and Kerns [4] have discussed extensively the accumulation of physical adsorption errors in the isothermal Burnett p – v – T apparatus. The best and most recent example of high-quality p – v – T measurements are those of Eubank [5] on butane with an isochoric Burnett apparatus. This apparatus couples isothermal Burnett measurements at supercritical temperatures (where adsorption is negligible) with isochoric measurements at lower temperatures, and thus does not allow the accumulation of adsorption errors.

Eubank et al. [6] have also examined at length random and systematic errors in sonic velocity and Joule–Thomson measurements, specifically as they are used to determine density second virial coefficients. Although these methods are not affected by adsorption, they do not directly measure B , and therefore a model has to be assumed for the temperature dependence of B so that the measurements can be reduced to B .

In the case of the sonic velocity, W , we have

$$W^2 = \frac{RT\gamma^{\text{id}}}{M} \left[1 + \left\{ 2B + 2(\gamma^{\text{id}} - 1) \frac{dB}{dT} + \frac{(\gamma^{\text{id}} - 1)^2 T^2}{\gamma^{\text{id}}} \frac{d^2B}{dT^2} \right\} \frac{p}{RT} + \dots \right] \quad (1)$$

where γ^{id} is the ideal-gas heat capacity ratio and M is the molecular weight of the gas.

Even when the Joule–Thomson coefficient data are properly extrapolated to $p \rightarrow 0$ (or $v \rightarrow \infty$), by plotting $\mu_{\text{JT}} C_p$ versus pressure and taking the zero-pressure intercept, we are still left with the relationship.

$$\mu_{\text{JT}} = - \frac{1}{C_p^{\text{id}}} \left[B - T \frac{dB}{dT} \right] \quad (2)$$

where C_p^{id} is the ideal-gas isobaric heat capacity. Eubank et al. [6] recommend that Joule–Thomson data be integrated numerically to obtain B because of the major undetectable errors caused by the need to assume a temperature model.

2.2. Data analysis

Dymond et al. [1] represented the temperature dependence of B with the equation

$$B/\text{cm}^3 \text{ mol}^{-1} = \sum_{i=0}^{i=N} a_i (298.15/T - 1)^i \quad (3)$$

As noted in the 1986 review, a satisfactory fit can be obtained with only four terms in Eq. (3) (up to $i = 3$). The fit of the data with Eq. (3) is discussed in Section 3.

The data, especially those for the alkanols, the alkyl ethers, and all the binaries, have also been analyzed with the 1974 correlation of Tsonopoulos [3]. This correlation is not described here, but a full description can be found either in the original paper [3] or in the 1989 review [2].

3. C₄–C₈ normal alkanes

Although the focus here is on C₄ + normal alkanes, it is worth noting recent reviews of the data for methane (Friend et al. [7]; Setzmann and Wagner [8]) and ethane (Friend et al. [9]). Our 1986 recommendations can still be used for propane.

New recommendations are made for butane, but no changes are made to the values we recommended in 1986 and 1989 for pentane through octane. The new data for butane are also compared with three well-known correlations.

3.1. Butane

In 1989 we modified the 1986 recommendations on the basis of the sonic velocity measurements of Ewing et al. [10] at 250–320 K, which were more positive than the 1986 recommendations. The more recent data of Ababio et al. [11] at 318.15–348.15 K are in excellent agreement with the 1986 recom-

Table 1

Second virial coefficient of *n*-butane: Comparison of 1986 recommendations [1] and Ewing's [10] results with Eubank's [5] measurements

<i>T</i> /K	<i>B</i> /cm ³ mol ^{−1} [Eubank]	(<i>B</i> − <i>B</i> ₈₆)/cm ³ mol ^{−1}	(<i>B</i> − <i>B</i> [Ewing])/cm ³ mol ^{−1}
264.9	−957.6	41.0	−22.5
275	−871	33	−17.4
300	−706	18	−10.2
325	−586.8	9.2	−5.2
350	−494.5	6.6	1.3 extr.
375	−426.5	1.8	3.0 extr.
400	−369.1	1.4	7.8 extr.
425	−321.5	1.8	12.9 extr.
450	−281.5	2.2	17.9 extr.

recommendations (average deviation of 3.8 cm³ mol^{−1}) and are 20 cm³ mol^{−1} more negative than Ewing's value at 320 K.

The picture for butane changed drastically with the very recent isochoric Burnett measurements of Eubank [5]. These measurements are accepted as our recommendations for butane. As shown in Table 1, they are significantly more positive than the 1986 recommendations below 350 K (Eubank noted significant adsorption below 375 K), in agreement with Ewing's observation that the low-temperature data in the literature are too negative due to adsorption errors. On the other hand, however, Eubank's measurements are more negative than Ewing's measurements up to 320 K—and more positive than Ewing's extrapolations above 320 K, where the 1986 recommendations are confirmed (especially above 350 K) by Eubank's measurements. The coefficients of Eq. (3) are given in Table 2. Because Eq. (3) is unsuitable for extrapolation, Table 2 also includes the temperature range of applicability of the coefficients.

As noted by Eubank, the temperature range of Ewing's measurements, 250–320 K, is too narrow to allow the reliable calculation of dB/dT and d^2B/dT^2 , which are used in the identity connecting the sonic velocity to the density *B* (see Eq. (1)).

Table 2

Coefficients of Eq. (3) (with applicable temperature range)

Compound	<i>a</i> ₀	<i>a</i> ₁	<i>a</i> ₂	<i>a</i> ₃	(<i>T</i> /K)
<i>n</i> -Butane	−716.138	−1699.734	−1622.427	−1191.374	(264–450)
<i>n</i> -Heptane	−3027.255	−10100.571	−14204.230	−8648.394	(313–623)
<i>n</i> -Octane	−3809.205	−10446.986	−9983.668	−3924.205	(337–575)
Methanol	−1935.566	−11412.862	−26595.565	−22266.860	(280–623)
Ethanol	−2842.871	−17213.987	−40628.892	−34501.245	(313–573)
1-Propanol	−2839.119	−14743.397	−31539.079	−25219.600	(378–573)
1-Butanol	−3377.964	−12935.865	−13996.315	—	(393–573)
1-Hexanol	−1613.848	−2235.066	—	—	(473–610)
Dimethyl ether	−468.217	−1287.717	−2523.350	—	(273–373)
Diethyl ether	−1238.059	−4311.055	−5782.276	—	(280–434)
Diisopropyl ether	−2051.611	−7072.101	−16280.838	—	(308–342)
Di- <i>n</i> -propyl ether	−1939.828	—	12055.598	—	(308–363)

While Eubank's measurements have raised questions about the 1986 recommendations below 350 K and Ewing's measurements and extrapolations, they strongly support the "old" measurements of Connolly [12] at 344.26–444.26 K: the average deviation of Connolly's data from Eubank's recent measurements is $1.1 \text{ cm}^3 \text{ mol}^{-1}$ (with a maximum deviation of $2.3 \text{ cm}^3 \text{ mol}^{-1}$ at 344.26 K). Thus, Connolly's data are once again demonstrated to be of high quality, a point to be remembered when we review the data for octane.

3.2. Pentane

A slight modification of the 1986 recommendations was proposed in 1989, on the basis of unpublished data by Ewing at 260–320 K, which were in excellent agreement with those published (Ewing et al. [13]: 270–330 K). The 1989 recommendations at 280 and 290 K were approximately $10 \text{ cm}^3 \text{ mol}^{-1}$ more negative than Ewing's values, which were based on sonic velocity measurements. As already noted, Ewing's values for butane were also found to be too positive.

Finally, the data of An et al. [14] at 308.2–348.2 K are in excellent agreement with the 1986/1989 recommendations.

3.3. Hexane

An et al. [14] also investigated hexane at 323.2–373.2 K, but their values are more negative than the 1989 recommendations and are closer to the 1986 values. However, the 1989 recommendations were based on the then recent data of Spiske and Gaube [15] and selected earlier data.

Bich et al. [16] reported values over the wide temperature range 348–600 K. Bich's values are in excellent agreement with the 1986/1989 recommendations up to 575 K; their value at 599.82 K is $9 \text{ cm}^3 \text{ mol}^{-1}$ more positive than our recommendation. The very recent results of Abdulagatov et al. [17] at 523.15–623.15 K similarly are in excellent agreement with the 1986 recommendations, but their value at 663.15 K is about $35 \text{ cm}^3 \text{ mol}^{-1}$ (or 14%) more positive and may be in error.

3.4. Heptane

Fig. 1¹ shows the deviation of recent (An et al. [14]; Millat et al. [18]) and older data selected in 1986 from the fit with Eq. (3); the coefficients are given in Table 2.

Millat's values above 473 K are more negative than all other literature data. As a result of Millat's data, the fit with Eq. (3) (and the coefficients in Table 2) is $10\text{--}15 \text{ cm}^3 \text{ mol}^{-1}$ more negative for $T > 480 \text{ K}$ than the 1986 recommendations. However, as the following discussion on octane will reinforce, Millat's values above 473 K are considered as questionable.

3.5. Octane

The most extensive recent data for octane are those of Millat et al. [19]. As shown in Fig. 2, Millat's data are much more negative than those selected in 1986. In particular, they are 25–

¹ The data sources in the figures are identified by the year—first three letters of first author/first three letters of second author. Those references not given in this paper can be found in [1], [2], or [3], as indicated in the figure captions.

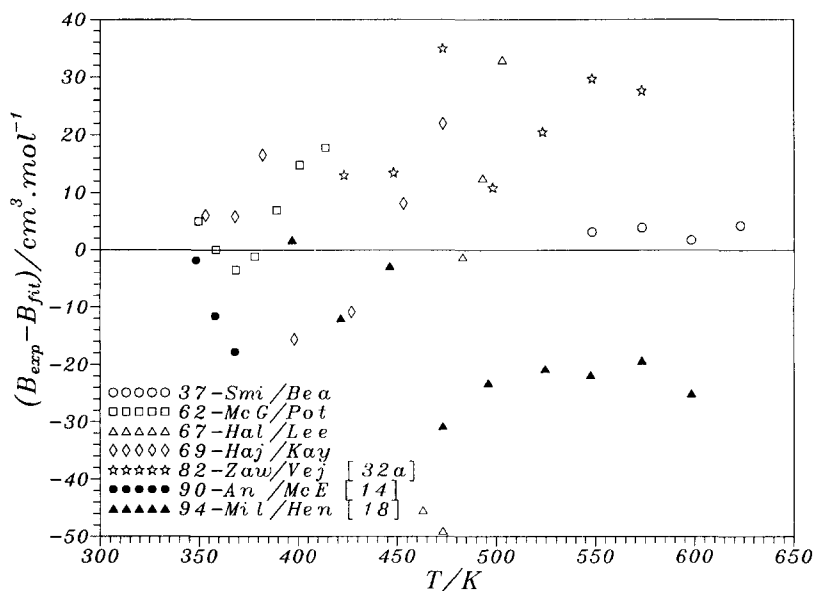


Fig. 1. Deviation of second virial coefficient data for *n*-heptane from Eq. (3). Dymond et al. [1] examined the data from the 1937–1982 references.

$30 \text{ cm}^3 \text{ mol}^{-1}$ more negative than the highly regarded measurements of Connolly and Kandalic [20]. (It was noted earlier that Eubank's [5] measurements on butane are in excellent agreement with Connolly's [12] data: average deviation between the two sets is $1.1 \text{ cm}^3 \text{ mol}^{-1}$.) Accordingly, our

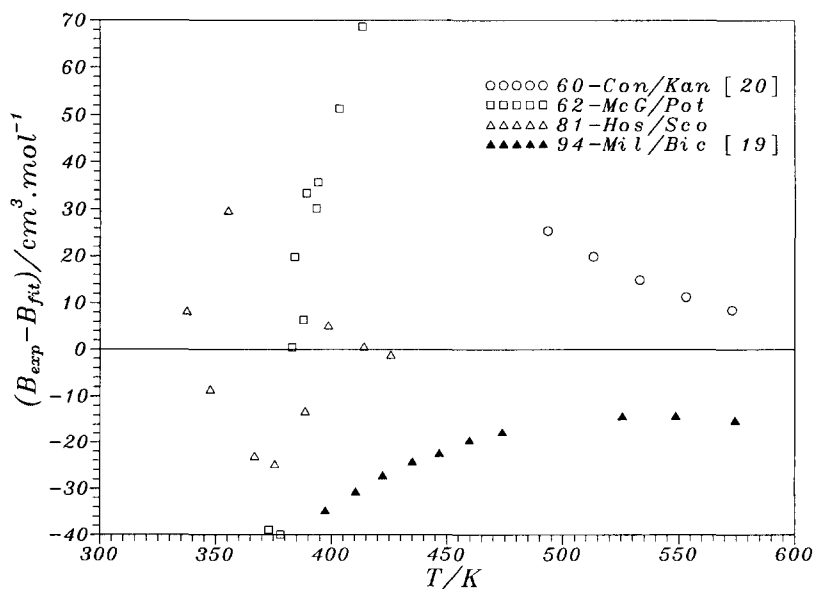


Fig. 2. Deviation of second virial coefficient data for *n*-octane from Eq. (3). Dymond et al. [1] examined the data from the 1960–1981 references.

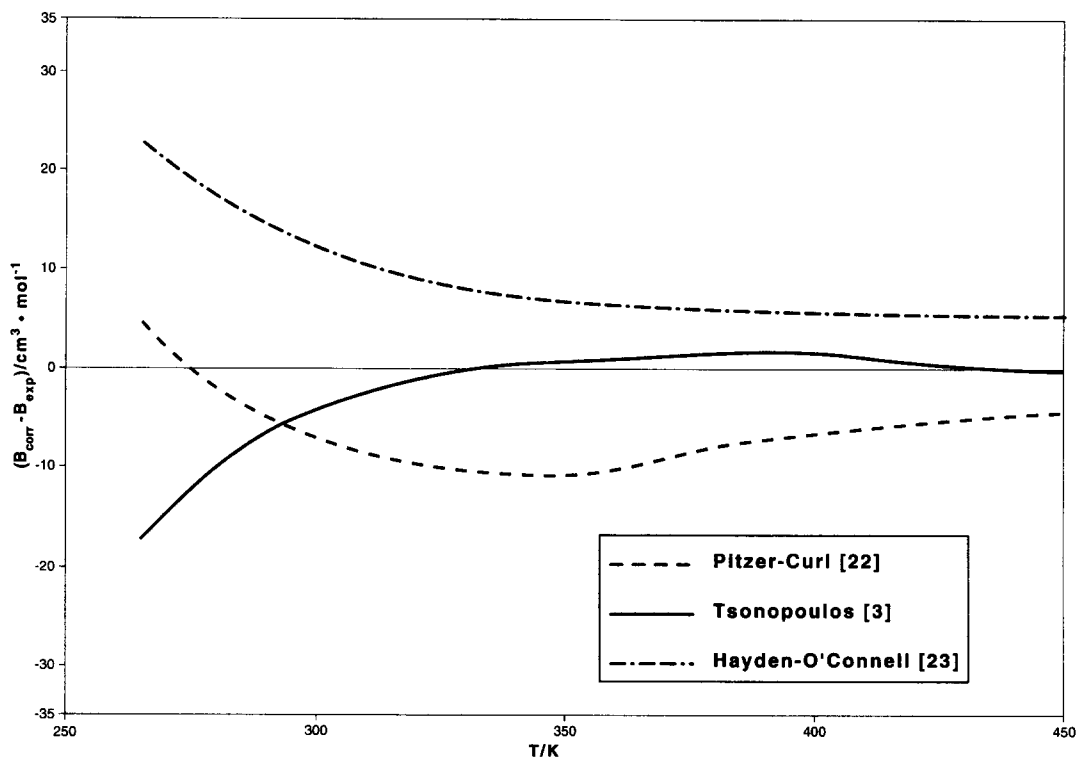


Fig. 3. Deviation of three correlations from Eubank's ([5]; see Table 1) data for *n*-butane; correlation values were taken from Cholinski et al. [21].

tentative recommendation is not to accept Millat's 1994 results for octane—and heptane—and instead keep our 1986 recommendations. Extrapolation of these recommendations to 623.15 K gives a value that is about $10 \text{ cm}^3 \text{ mol}^{-1}$ (or 2%) more negative than the result of Abdulagatov et al. [17].

In summary, the only modification we propose to our 1986/1989 recommendations for C_4 + normal alkanes is for butane, for which we accept Eubank's [5] isochoric Burnett measurements from 265 to 450 K as the best available data—along with Connolly's [12] p - v - T data at 344.26–444.26 K.

3.6. Comparison of Eubank's data with correlations

Fig. 3 plots the difference between three well-known correlations and Eubank's [5] data for butane. The values calculated with the correlations were taken from Cholinski et al. [21].

As can be seen in Fig. 3, the Pitzer–Curl² [22] correlation gives the best agreement with the data at 265 and 275 K, but at $T > 300 \text{ K}$ ($T_r \geq 0.7$) its modification proposed by Tsonopoulos [3] gives the best fit. The Hayden–O'Connell [23] correlation is everywhere too positive, but is slightly better than Pitzer–Curl [22] in the range $320 < T/\text{K} < 390$.

² Robert F. Curl, 1996 Nobel Prize winner in chemistry for discovery of fullerenes.

Thus, on the basis of the high-quality data for butane, it can be concluded that the modified Pitzer–Curl correlation [3] is the method of choice for $T_r \geq 0.7$, but the original correlation is superior at lower reduced temperatures.

4. C_1 – C_6 linear 1-alkanols

In 1989 we examined all available data and made preliminary recommendations for the first four alkanols. (The vapor pressure and enthalpy of vaporization data of Counsell et al. [24] for 1-pentanol were discarded as a source of B values.) Here we update our recommendations by examining more recent data, nearly all of them for methanol. A noteworthy addition is the work of Shakhverdiev et al. [25], the only known source of B data for 1-hexanol.

The data have been regressed both with Eq. (3) and with the 1974 correlation of Tsonopoulos [3], where the new values for the parameter b are compared with the 1974 results. The analysis with the correlation augments the data smoothing with Eq. (3) and makes it possible to test data for internal consistency.

At present, definitive recommendations can be made only for methanol. Also of interest is that the new data for 1-hexanol appear to support the relationship between b and the reduced dipole moment of the alkanol that was proposed in 1974.

4.1. Methanol

The deviations of the data selected in 1989 and of the more recent results from the fit with Eq. (3) are plotted in Fig. 4. There are very significant deviations below 375 K, but the data smoothing raises

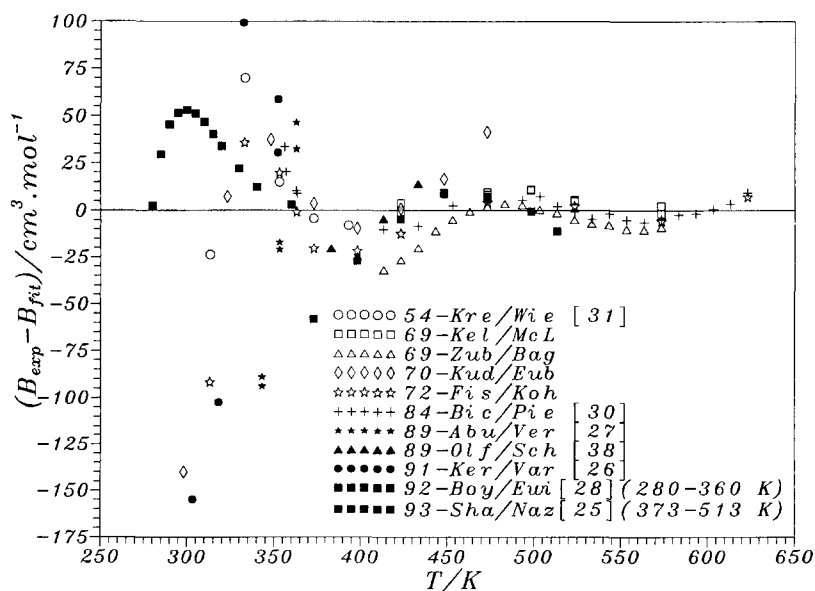


Fig. 4. Deviation of second virial coefficient data for methanol from Eq. (3). Tsonopoulos et al. [2] examined the data from the 1954–1984 references.

questions at high temperatures. The smoothed $\Delta B/\Delta T$ above 500 K exhibits no decreasing trend, and the fit is incorrect above 600 K (B at 623 K is more negative than at 613 K). These observations suggest that some of the recent data may have to be evaluated more closely.

Of the new results, those of Kerl and Varchmin [26], which apparently were based entirely on optical measurements, are the most questionable: they deviate from -150 to $+100 \text{ cm}^3 \text{ mol}^{-1}$ from the fit with Eq. (3). Also questionable, perhaps, are the Burnett measurements of Abusleme and Vera [27] at 343–363 K: -90 to $+40 \text{ cm}^3 \text{ mol}^{-1}$ off. Finally, the temperature dependence of the values of Boyes et al. [28], the only source of data below 300 K, apparently is significantly different from that of Eq. (3).

The recommendations made by de Reuck and Craven [29] in the recent IUPAC volume should also be noted. The IUPAC equation of state is based on the data of Bich et al. [30], which go down to 356 K. Extrapolation of the IUPAC equation below 350 K is *much* more positive than Boyes et al. [28]—whose data, in turn, are more positive than other data plotted in Fig. 3.

4.2. Ethanol

The only recent data for ethanol are those of Abusleme and Vera [27] at 343.15–363.15 K. As shown in Fig. 5, they differ greatly from the measurements of Kretschmer and Wiebe [31], and even those of Bich et al. [32].

The data smoothing with Eq. (3) gives a constant $\Delta B/\Delta T$ above 500 K. As a result, the smoothed B value at 580 K (which is in excellent agreement with the data of Bich et al. [32]) is more positive than that of methanol. These questions over the high-temperature B of ethanol were also raised in the 1989 review [2].

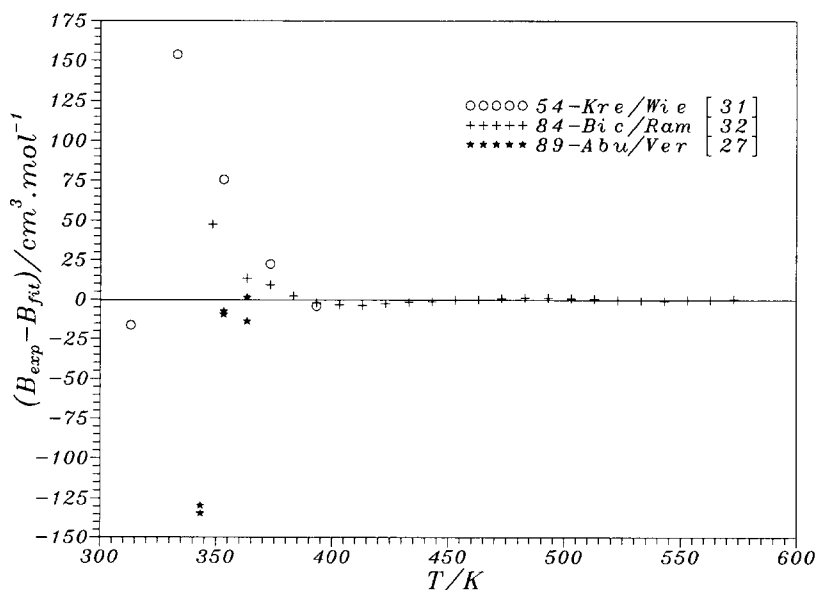


Fig. 5. Deviation of second virial coefficient data for ethanol from Eq. (3).

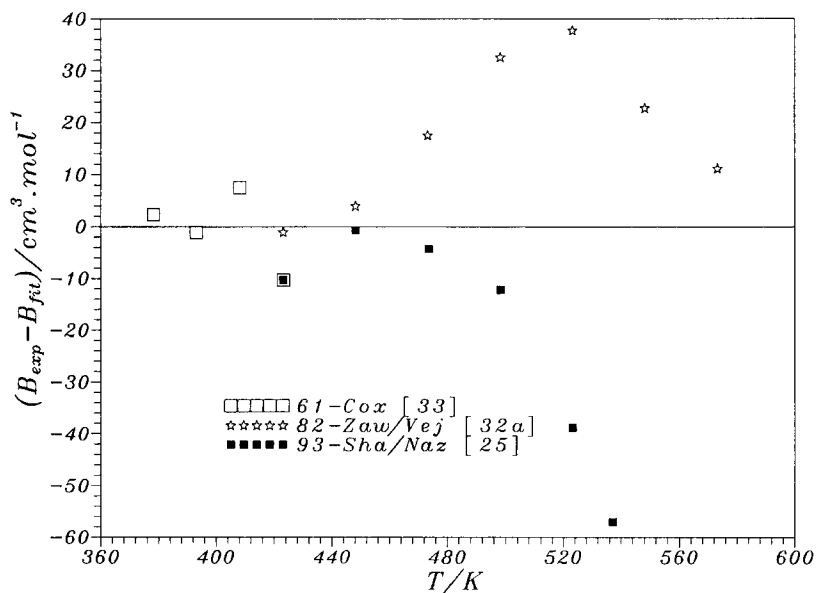


Fig. 6. Deviation of second virial coefficient data for 1-propanol from Eq. (3).

4.3. 1-Propanol, 1-butanol, and 1-hexanol

Figs. 6 and 7 show that the recent data of Shakhverdiev et al. [25] are significantly more negative than those of Zawisza and Vejrosta [32], for 1-propanol, and of Cox [33], for 1-butanol. In the case of 1-hexanol, there are no other data for comparison. In this case, a more meaningful comparison may be provided by using the 1974 correlation [3] to examine the data for the alkanols.

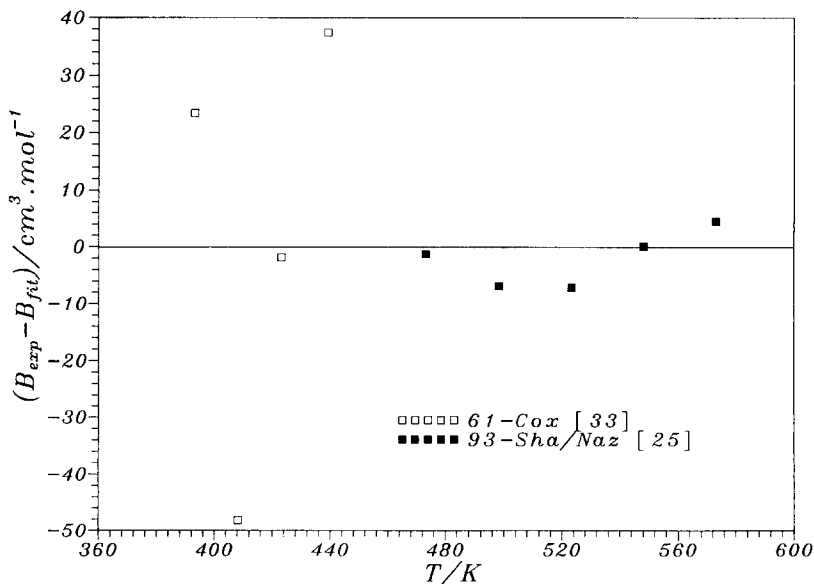


Fig. 7. Deviation of second virial coefficient data for 1-butanol from Eq. (3).

Table 3
Optimum and predicted b values for 1-alkanols in Eq. (4)

Alkanol (no. points)	1974 b	New b
Methanol (112)	0.0560	0.0525
Ethanol (34)	0.0572	0.0578
1-Propanol (17)	0.0447	0.0461
1-Butanol (9)	0.0367	0.0408
1-Hexanol (7)	0.0273 (Eq. (5))	0.0250

4.4. Comparisons with the 1974 correlation

In the 1974 correlation, the polar contribution to the B of 1-alkanols is given by

$$f^{(2)} = 0.0878/T_r^6 - b/T_r^8 \quad (4)$$

where the parameter b for $C_2 + 1$ -alkanols is expressed as a function of the alkanol's reduced dipole moment, μ_r :

$$b = 0.00908 + 0.0006957\mu_r \quad (5)$$

where $\mu_r = 10^5 \mu^2 p_c / T_c^2$; μ in Debyes, p_c in atm, T_c in K.

Table 3 presents the optimum b values determined in 1974 [3]; those calculated with Eq. (5); and, finally, the new results. The table also gives the number of B data points used in this work.

The new b values agree closely with the 1974 values (or, in the case of 1-hexanol, the value calculated with Eq. (5)). In view of the extensive database for methanol, the 1974 recommendation, 0.056, should be replaced by the new value: $b = 0.0525$.

The optimum value for 1-hexanol, $b = 0.025$, is in satisfactory agreement with the value 0.0273 estimated with Eq. (5)—or the value 0.03 estimated by Bo et al. [34]. Thus, it appears that the data of Shakhrediev et al. [25] for 1-hexanol are internally consistent with what is known for the lower 1-alkanols. Unfortunately, however, the database for $C_2 + 1$ -alkanols is too limited to support any definitive recommendations for their second virial coefficients.

5. Water

The 1974 analysis of the data for water [3] was based entirely on Kell's [35] data below 450°C and recommended that the polar term in the 1974 correlation be given by

$$f^{(2)} = 0.0279/T_r^6 - 0.0229/T_r^8$$

although it was noted that a nearly equally good fit could be provided with a one-parameter $f^{(2)}$. Unfortunately, Kell's [35] data have been shown by Eubank et al. [36] and others to be too negative due to adsorption effects that had not been accounted for, and therefore the 1974 analysis had to be revised.

Tsionopoulos and Heidman [37] reanalyzed in 1990 all the data in Cholinski et al. [21], along with Sets II and III of Eubank et al. [36], and recommended a simpler $f^{(2)}$:

$$f^{(2)} = -0.0109/T_r^6$$

The revised recommendation fits all 118 data points with an average deviation of $11.0 \text{ cm}^3 \text{ mol}^{-1}$.

Abdulagatov et al. [17] reported new data for water at 523.15–653.15 K. These data are about $5 \text{ cm}^3 \text{ mol}^{-1}$ more positive than the values predicted with the revised correlation, which is consistent with what is shown in Fig. 3 of Tsonopoulos and Heidman [37] for other literature data.

6. Alkyl ethers

Very limited work on the B of alkyl ethers has been done since the analysis of Tsonopoulos [3]. The only known significant addition is the investigation of Olf et al. [38] on diethyl ether. As a result, our conclusions are similar to those drawn in 1974. Adequate data are available only for the dimethyl and diethyl ethers, and to a much more limited extent for diisopropyl ether.

Fortunately, alkyl ethers are very weakly polar. Indeed, the polar contribution to B for ethers heavier than diethyl ether can be neglected—although such a contribution may be important in predicting the cross-coefficient of ether binaries with polar compounds. This is discussed in Section 7.

The weak polarity of alkyl ethers also means that the lack of B data for the gasoline additives (MTBE, TAME, ETBE, TAEE) is not an important concern. The recommendations given for the ethers used or considered as gasoline additives include some recent determinations of their critical properties that were sponsored by DIPPR®.

6.1. Dimethyl ether

Fig. 8 is a plot of the deviations of the experimental data for DME from the fit with Eq. (3). Apparently no data have been reported since 1971. While the deviations are relatively small, the high-temperature data of Osipiuk and Stryjek [39] are most likely too negative. This apparent

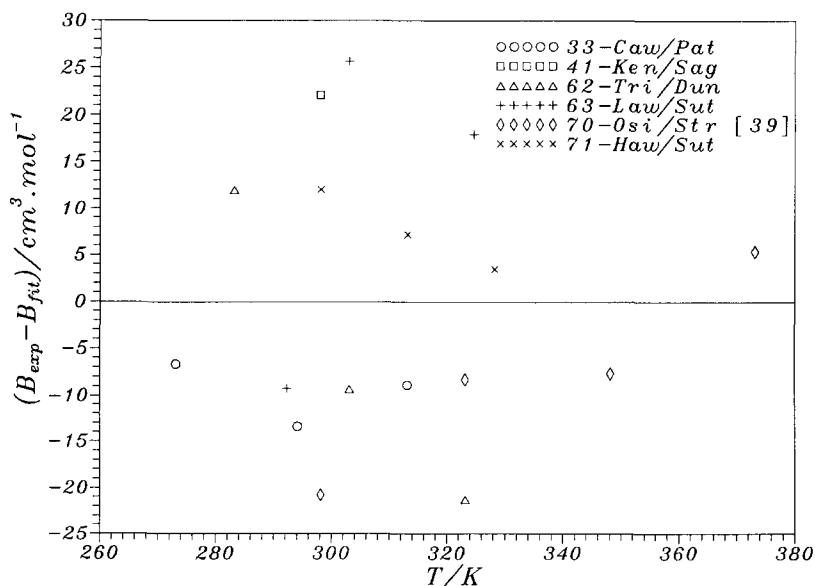


Fig. 8. Deviation of second virial coefficient data for dimethyl ether from Eq. (3). Tsonopoulos [3] examined the same data.

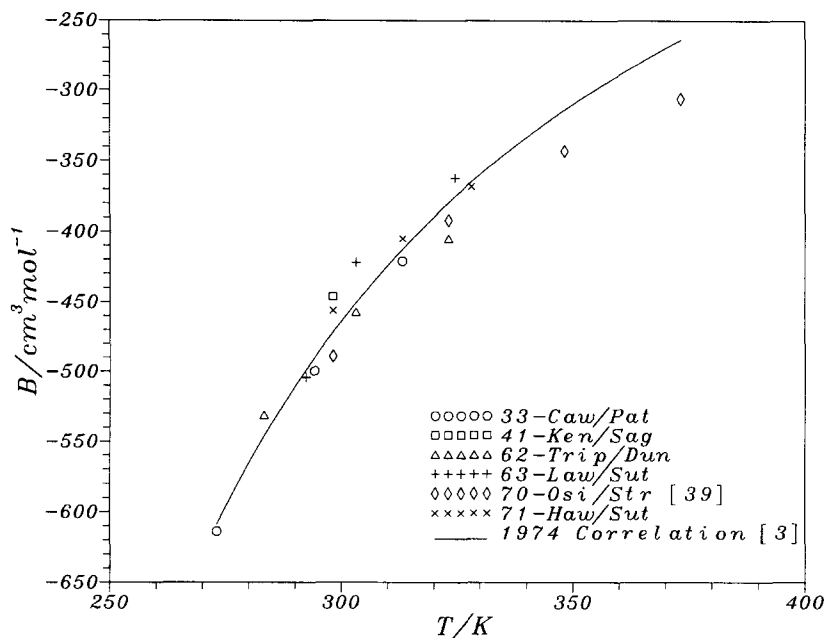


Fig. 9. Comparison of second virial coefficient data for dimethyl ether with the 1974 correlation [3]. Tsonopoulos [3] examined the same data.

discrepancy is more clearly illustrated in Fig. 9, where the 1974 correlation, which is considered to be reliable at high (reduced) temperatures, is much more positive than the data of Osipiuk and Stryjek [39].

6.2. Diethyl ether

As in the case of dimethyl ether, the deviation of the data from Eq. (3) in Fig. 10 does not provide as clear a picture as Fig. 11, where the fit with the 1974 correlation demonstrates that the relatively recent data of Olf et al. [38] probably are of higher quality than those, for example, of Lambert et al. [40]. In the 1974 analysis it had been noted that the data of Lambert et al. [40] are consistently too positive. Fig. 11 supports that observation—as well as that the data of Knoebel and Edmister [41] have too steep a temperature dependence.

6.3. Diisopropyl ether

In 1974, on the basis of Chevalier's [42] data, it was noted that neglecting the polar contribution degraded the fit only slightly. The data of Andon et al. [43] are in relatively good agreement with Chevalier's, as shown in Fig. 12, and thus the 1974 conclusion is still applicable.

6.4. Di-n-propyl ether

The satisfactory agreement of Andon's data for DIPE with Chevalier's is not observed for DNPE. Fig. 13 clearly illustrates the gross disagreement between Andon et al. [44] and Chevalier [42]. The

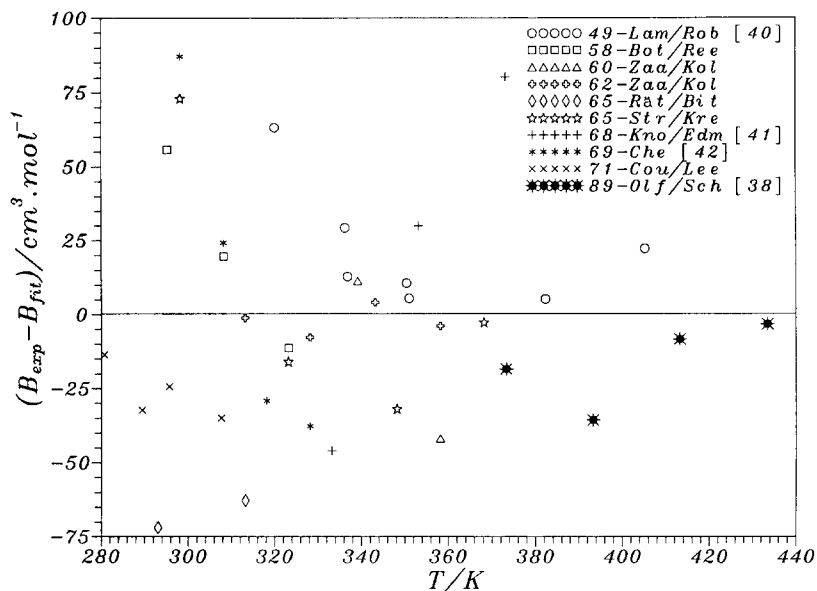


Fig. 10. Deviation of second virial coefficient data for diethyl ether from Eq. (3). Tsionopoulos [3] examined the data from the 1949–1971 references.

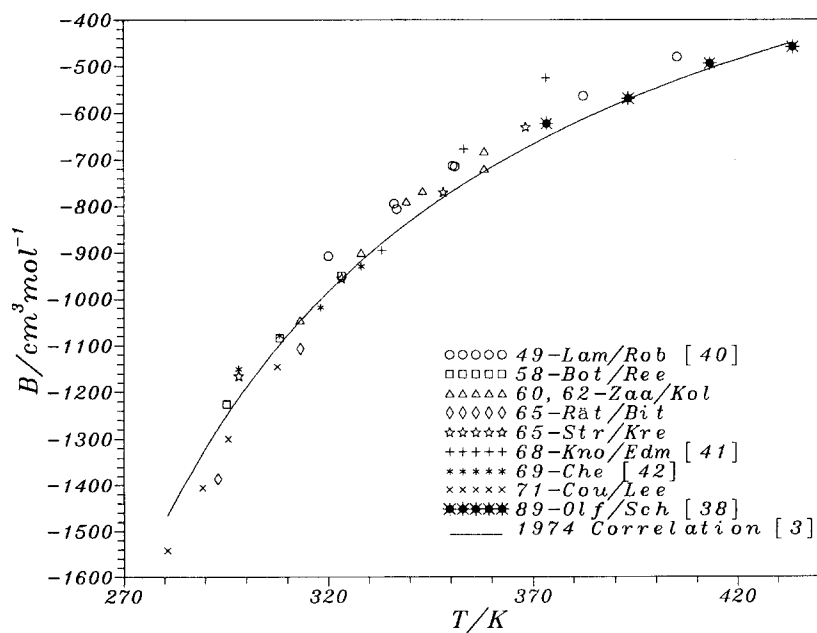


Fig. 11. Comparison of second virial coefficient data for diethyl ether with the 1974 correlation [3]. Tsionopoulos [3] examined the data from the 1949–1971 references.

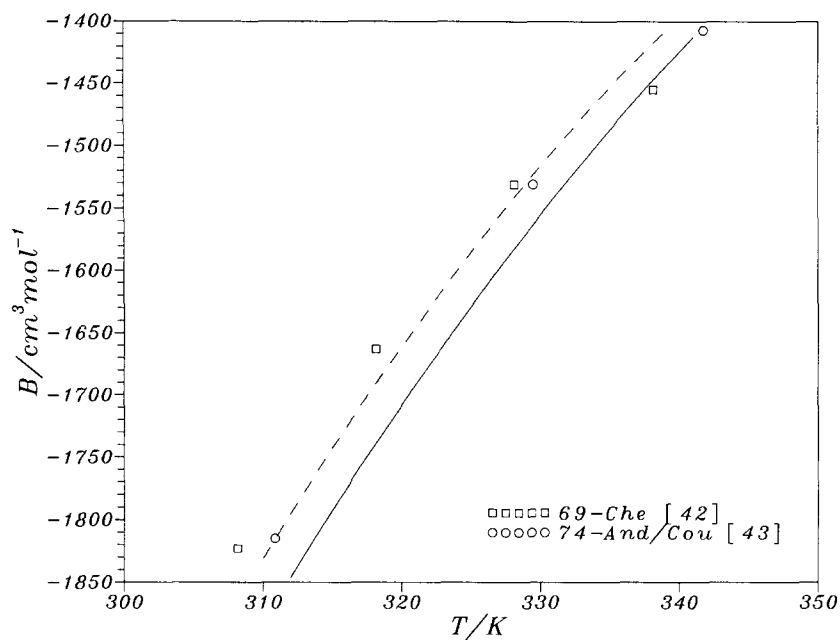


Fig. 12. Comparison of second virial coefficient data for diisopropyl ether with the 1974 correlation [3]: —, with polar contribution; ---, nonpolar correlation.

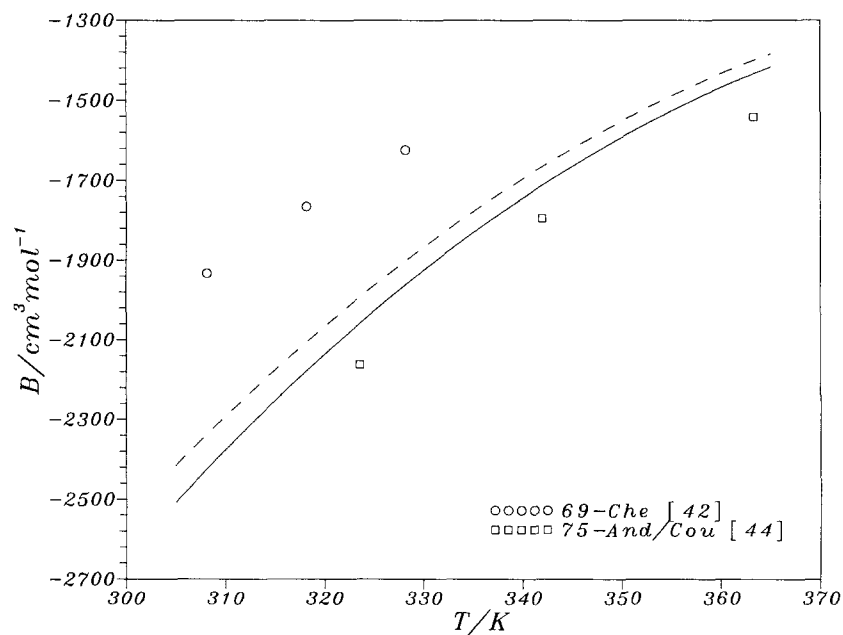


Fig. 13. Comparison of second virial coefficient data for di-*n*-propyl ether with the 1974 correlation [3]: —, with polar contribution; ---, nonpolar correlation.

Table 4

Comparison of other alkyl ether data with 1974 correlation [3]

	Avg. Dev.	Bias (data - corr.)	
Osipiuk and Stryjek [39]			
dimethyl ether	32.3	– 32.3	
methyl ethyl ether	138	138	too positive?
methyl- <i>n</i> -butyl ether	191	– 191	too negative?
methyl- <i>n</i> -pentyl ether	41.2	– 6.8	
Chevalier [42]			
diethyl ether	28.2	16.3	
ethyl- <i>n</i> -propyl ether	139	139	too positive?
ethyl- <i>n</i> -butyl ether	471	471	too positive?
diisopropyl ether	59.3	55.4	
di- <i>n</i> -propyl ether	414	414	too positive?
di- <i>n</i> -butyl ether	895	895	too positive?

latter data are probably too positive, and those of Andon et al. [44] are too negative, as will be discussed next.

6.5. Other alkyl ethers

Osipiuk and Stryjek [39] and Chevalier [42] have measured the B of several other ethers. These are listed in Table 4, along with the average deviation and bias (data - correlation) resulting from comparing the data with the 1974 correlation.

In the case of Osipiuk and Stryjek [39], it has already been noted that their high-temperature data for DME are most likely too negative. More striking is the apparent discrepancy between their data for methyl ethyl ether (more positive than predictions with $f^{(2)} = 0$) and methyl butyl ether (too negative?).

Chevalier's [42] data are consistently too positive, especially for the ethyl-*n*-propyl, ethyl-*n*-butyl, di-*n*-propyl, and di-*n*-butyl ethers. For all these ethers, Chevalier's data are *much* more positive than predictions with $f^{(2)} = 0$. However, the 1974 nonpolar correlation is known to be fairly reliable, especially for $T_r > 0.7$ (see earlier discussion on butane).

In the case of DNPE, the 1974 nonpolar correlation ($f^{(2)} = 0$) predicts values that roughly are half way between the data of Chevalier [42] and of Andon et al. [44]. Fitting the latter data would require a much more negative $f^{(2)}$ than that for DIPE or that estimated from the 1974 relationship between a and μ_r , the reduced dipole moment of the alkyl ethers.

6.6. Polar contribution to B of alkyl ethers

Table 5 lists the optimum values for the parameter a that were determined in 1974 and in the present study. In addition, it lists the values predicted with a general relationship between a and μ_r for ketones, aldehydes, and alkyl ethers (and acetonitrile):

$$a = -2.14 \times 10^{-4} \mu_r - 4.308 \times 10^{-21} \mu_r^8 \quad (6)$$

Table 5
Optimum and predicted a values for alkyl ethers

Alkyl ether (no. points)	1974 a	New a	Eq. (6)	Eq. (7)
Dimethyl ether (17)	−0.01513	−0.01524	−0.01198	−0.01513
Diethyl ether (37)	−0.00202	−0.00450	−0.00484	−0.00225
Diisopropyl ether (7)	−0.00089	−0.00155	−0.00310	−0.00089

The new and the 1974 optimum value for DME are essentially the same, but the new data for DEE have led to a more negative value for a , which is closer to that predicted by Eq. (6). The addition of the data of Andon et al. [43] for DIPE has also made its new a more negative.

In addition to the general relationship (6), Tsonopoulos [3] also proposed a special relationship for alkyl ethers:

$$\ln(-a) = -12.63147 + 2.09681 \ln \mu_r \quad (7)$$

As shown in Table 5, Eq. (7) is closer to the optimum a values for DME and DIPE than is Eq. (6). But the data for DEE, and to a lesser extent those for DNPE (Andon et al. [44]) and methyl-*n*-butyl ether, are closer to Eq. (6).

The available data for alkyl ethers are too limited in quantity and quality to support definitive recommendations. The new a values in Table 5 for DME and DEE are tentatively recommended for calculations with the 1974 correlation, but more data are needed to clarify the situation for other alkyl ethers. For those others, the general relationship (6) is recommended—until more data can determine whether Eq. (6) or Eq. (7) should be used for alkyl ethers.

6.7. Recommendations for gasoline additives

Table 6 presents the critical properties (some of them based on very recent measurements) of alkyl ethers that are used or considered as gasoline additives: MTBE, TAME, ETBE, DIPE, and TAEE. Also included in the table are values for their acentric factor and the parameter a (calculated with Eq. (6)). As previously noted, the a values are too small (in absolute magnitude) to have a significant effect on the pure B , but may affect the prediction of B_{ij} ($i \neq j$) for ether (i)/polar (j) binaries. This is because $a_{ij} \neq 0$ only when *both* i and j are polar.

Table 6
Recommendations for gasoline additives

	T_c /K	p_c /MPa	ρ_c /g cm ^{−3}	ω	a
MTBE ^a	497.1	3.430	0.27	0.266	−0.0044
TAME ^b	536.2	3.191	0.271	0.300	−0.0032
ETBE ^b	509.4	2.934	0.259	0.337	−0.0034
DIPE ^a	500.3	2.832	0.265	0.332	−0.00155
TAEE ^c	546	2.935	0.26	0.367	−0.002

^a Ambrose et al. [45]

^b DIPPR[®] 851 measurements: Wilson et al. [46]

^c DIPPR[®] 821 measurements and calculations: Steele et al. [47]; the results for TAME in Ref. [47] are close to those in Ref. [46].

7. Cross-coefficients

The reliable prediction of the B of mixtures depends on how accurately we can estimate, for each binary, the second virial cross-coefficient, B_{ij} ($i \neq j$). In the 1974 correlation (and, generally, any correlation), the key mixing rule for the prediction of B_{ij} is that for the characteristic critical temperature T_{cij} , which is given by

$$T_{cij} = (T_{ci}T_{cj})^{0.5}(1 - k_{ij}) \quad (8)$$

Thus, the first step in the data analysis of mixture B or B_{ij} (which can be determined directly by measuring the solubility of a condensed component i in compressed gas j) is to determine the optimum value of k_{ij} for each binary.

In addition to mixing rules for p_{cij} and ω_{ij} (see Tsonopoulos [3]), the 1974 correlation requires also mixing rules for a_{ij} and b_{ij} . These polar parameters are set equal to zero even if one of the components is nonpolar. If *both* components are polar, then it is assumed that

$$a_{ij} = 0.5(a_i + a_j) \quad (9a)$$

$$b_{ij} = 0.5(b_i + b_j) \quad (9b)$$

The following review of data for alkane/alkanol/water/alkyl ether binaries focuses on the value of the k_{ij} parameter that can be determined by data regression—or estimated, in the absence of data. Tsonopoulos et al. [2] have also demonstrated that the k_{ij} of (dissimilar) alkane/alkane binaries can be predicted reliably from the pure-component critical volumes. The same approach may also be usable for alkanol/alkanol and ether/ether binaries.

7.1. 1974 recommendations

On the basis of the then limited database, Tsonopoulos [3] recommended the following rough guide for the binaries of interest.

	k_{ij}	(No. binaries)
alkanes/alkanols	0.15	(2)
alkanes/water	0.40	(2)
alkanes/ethers	0.10	(1)
alkanols/water	0.10	(0)
alkanols/ethers	0.13	(0)
water/ethers	0.35	(0)

The number of binaries on which these recommendations were based is given in parentheses. Since no data were available for the bottom three classes of binaries, estimates had to be provided so that the 1974 correlation could be used for these binaries.

More data were analyzed in later publications (Tsonopoulos [48]; Tsonopoulos et al. [2]; Tsonopoulos and Heidman [37]), but still no data have been found for any water/ether binaries.

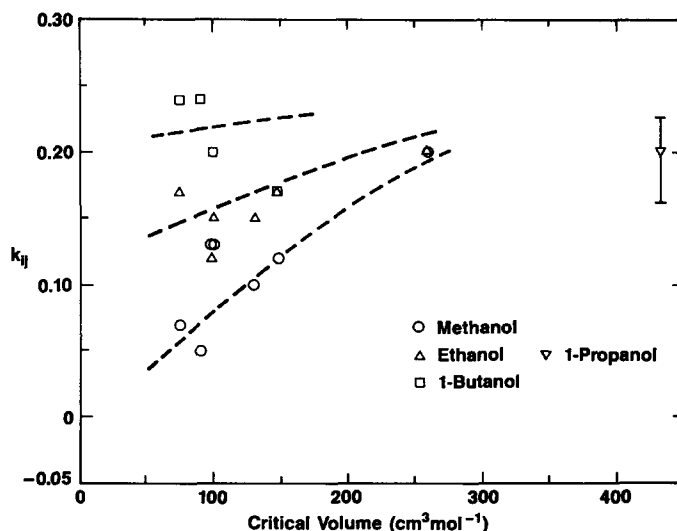


Fig. 14. Optimum k_{ij} values for nonpolar binaries of 1-alkanols; from Tsonopoulos et al. [2].

7.2. Alkanes / alkanols

An extensive analysis and correlation of the k_{ij} values for alkane/alkanol binaries was presented at the first IUPAC workshop on alkanes/alkanols and was documented by Tsonopoulos et al. [2]. Fig. 14 presents the data and a tentative correlation of k_{ij} with the critical volume of the hydrocarbon (or other nonpolar compound).

The only new data that have been located are those of Sauermann et al. [49] for hexane/ethanol at 473 and 507 K. Their Burnett measurements on a nearly equimolar mixture were reduced to the following B_{ij} values.

T/K	$B_{12}/\text{cm}^3 \text{ mol}^{-1}$
473.15	$-340 (\pm 30)$
507.15	$-280 (\pm 30)$

Data regression with the 1974 correlation gave

$$k_{12} = 0.1 \quad (\text{hexane/ethanol})$$

This value is lower than those for ethanol binaries in Fig. 14, but is not as discordant as some of the values reviewed and discarded by Tsonopoulos et al. [2].

7.3. Alkanes / water

Fig. 15, which is taken from Tsonopoulos and Heidman [37], shows a very strong correlation between k_{ij} and the critical volume of the hydrocarbon, which is simply given by

$$k_{ij} = 0.6114 - 2.7135/v_{c,j}^{0.5} \quad (i = \text{water}; j = \text{alkane})$$

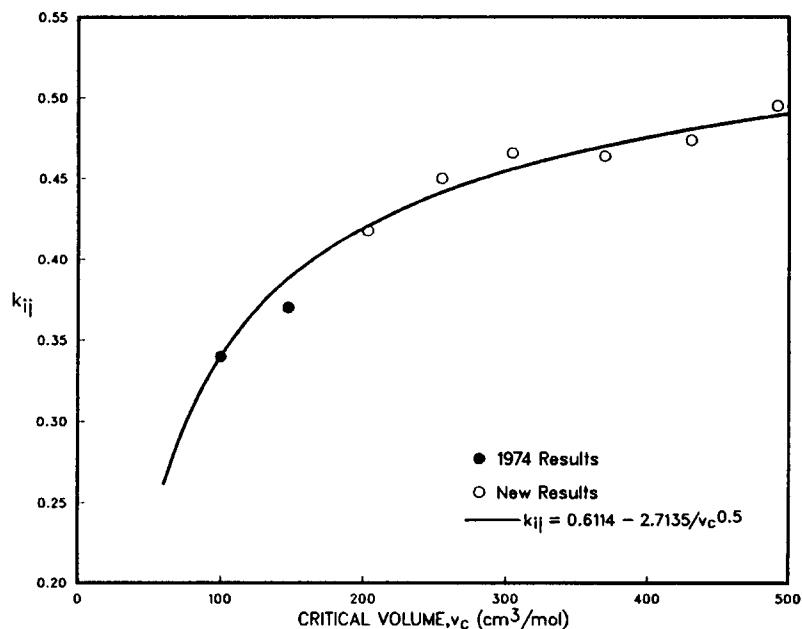


Fig. 15. Optimum k_{ij} values for n -alkane binaries of water; from Tsonopoulos and Heidman [37].

New data have been reported for binaries of water with methane (Abdulagatov et al. [50]) and hexane (Abdulagatov et al. [51]). Our calculations of B_{12} from the data for near-equimolar mixtures gave values that were significantly different from values reported in their figures. They are also different from the values recently reported in tabular form (Abdulagatov et al. [17]), which have been accepted as the final results from Abdulagatov's group. Ref. [17] also includes a single value for water with octane.

Fig. 16 plots all known B_{ij} data for methane/water (except for those of Aoyagi et al. [52] at 240–270 K). The 1974 analysis had been based on the data of Olds et al. [53] and of Rigby and Prausnitz [54]. The more recent data of Smith et al. [55] and of Joffrion and Eubank [56] are in excellent agreement with the 1974 recommendation of $k_{ij} = 0.34$ for methane/water. Analysis of all the data, including those of Abdulagatov et al. [17], but not the apparently erratic data of Skripka [57], led to a slightly different value: $k_{ij} = 0.32$. Abdulagatov's results for hexane/water and octane/water were also found to be consistent with the literature data examined by Tsonopoulos and Heidman [37].

7.4. Alkanes / ethers

The 1974 recommendation, $k_{ij} = 0.10$, was based on data for diethyl ether with hexane (0.08) and, in the absence of any other data with alkanes, with benzene (0.10). Tsonopoulos [48] presented additional results for diethyl ether with methane (0.12) and ethane (0.06), which generally support the 1974 recommendation.

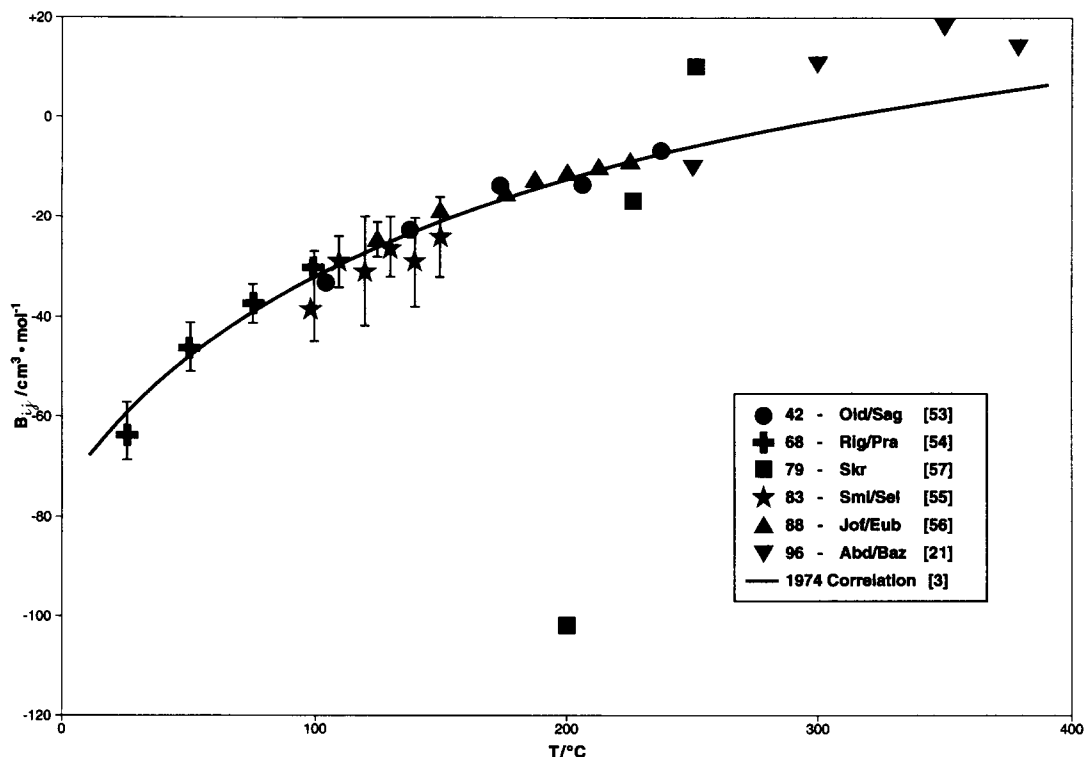


Fig. 16. Comparison of B_{ij} values for methane/water with 1974 correlation ($k_{ij} = 0.32$).

7.5. Alkanols / water

The only known data for alkanol/water binaries are those of Noppe et al. [58] for water with methanol and ethanol. Both binaries were investigated at 400–525 K, and the analysis with the 1974 correlation produced the following values:

$$k_{ij} = 0.0 \quad (\text{methanol/water})$$

$$k_{ij} = 0.05 \quad (\text{ethanol/water})$$

These values are accepted for the corresponding binaries, along with the 1974 rough estimate, $k_{ij} = 0.1$, for $C_3 +$ alkanols with water.

7.6. Alkanols / ethers

Olf et al. [38] have measured the B of methanol/diethyl ether at 398–433 K, the only known data for alkanol/ether binaries. A good fit of these data with the 1974 correlation requires the value

$$k_{ij} = 0.10$$

which should supersede the 1974 rough estimate of 0.13.

7.7. Water / ethers

In the absence of any data, the 1974 rough estimate of

$$k_{ij} = 0.35$$

can still be used, especially since it was based on the value 0.40 for water/alkanes, which is still supported by the more detailed results in Fig. 15.

8. Recommendations

The only definitive recommendation that can be made for normal alkanes is for butane, thanks to the recent data of Eubank [5]. These data also suggest that the 1974 correlation is reliable for $T_r > 0.7$, but is too negative at lower reduced temperatures.

For alkanols, it should be possible to make recommendations for methanol, preferably after the uncertainties of B values from p - v - T , sonic velocity, and Joule–Thomson measurements are reevaluated. If a “final” set of B values is selected, then the polar contribution to B in the 1974 correlations (and others?) can be reconsidered, especially for $T_r > 0.7$.

At present, the data for C_2 + linear 1-alkanols and alkyl ethers are probably insufficient in quality and temperature range to support any definitive recommendations. Experimenters are thus encouraged to make high-quality measurements on both pure C_2 + alkanols and alkyl ethers *and* their mixtures.

The need for a reliable correlation is especially clear for B_{ij} values, because it is impossible to measure all binaries of interest in vapor–liquid separations. The 1974 correlation, with the k_{ij} values that have been updated in this paper, may continue to be useful in VLE calculations.

9. List of symbols

a_i	coefficients of Eq. (3)
a, b	parameters of polar contribution term to B , $f^{(2)}$
B	second virial coefficient
C_p^{id}	ideal-gas isobaric heat capacity
$f^{(2)}$	polar contribution term to B
k_{ij}	characteristic binary constant; see Eq. (8)
M	molecular weight
p	pressure
R	gas constant
T	absolute temperature
v	molar volume
W	sound velocity

9.1. Greek letters

γ^{id}	ideal-gas heat capacity ratio
Δ	difference

μ	dipole moment
μ_{JT}	Joule–Thomson coefficient
ω	acentric factor

9.2. Subscripts

c	critical property
i, j	property of component i, j
ij	property of i – j interaction
r	reduced property

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